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22 May 1974
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15 June 1966

MILITARY SPECIFICATION

PROJECTILE, 105MM, SMOKE, WP, M60
METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, 105MM, Smoke, WP, M60 Metal Parts assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1459	- Macrograph Standards for Steel Bars, Billets and Blooms for Ammunition Components

FSC 1315

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DRAWINGS

Armament Command

D10535390	- Projectile, 105MM, Smoke, WP, M60 Metal Parts Assembly
LI 10535390	- Index to Inspection Equipment List for Projectile, 105MM, Smoke, WP, M60 Metal Parts Assembly
F8837835	- Pallet for Projectile, Empty or Loaded

PUBLICATIONS

Frankford Arsenal

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection

(Copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM E-8	- Tension Testing of Metallic Materials
ASTM E-10	- Brinnel Hardness Testing of Metallic Materials
ASTM E-117	- Salt Spray (FOG) Testing
ASTM A-317	- Macro-Etch Testing for Steel

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

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3.2 Bar or billet soundness.-Material supplied in accordance with the ASTM specifications referenced on the applicable drawing shall not have center defects greater than "A5" of MIL-STD-1459. Material supplied in accordance with the Military (MIL) Specifications references on the applicable drawing shall comply with the macro-etch requirements of the applicable specification.

3.3 Heat treatment or stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.3.1 Hot forge - heat treat method.-Prior to finish machining projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (F), followed by an oil quench. After quenching, the body forgings shall be tempered without delay at a temperature above 750°F suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700°F, after which cooling may be accelerated by water spray.

3.3.2 Cold shaping and hot cup - cold draw methods.-Cold working shall be so accomplished as to produce in the projectile body the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700°F, minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified on the applicable drawing. The temperature and time duration of a process anneal prior to final forming operation shall be established by the ability to meet the required mechanical properties when formed by an approved sequence of operations.

3.4 Mechanical properties.

3.4.1 Body.-Mechanical properties specified on the applicable drawing shall apply throughout the component.

3.4.2 Rotating band.-After application and subsequent removal the rotating band shall have the following mechanical properties:

Tensile Strength - 30,000 lbs. per square inch (psi) minimum
Elongation - 25 percent per inch of gage length, minimum

3.5 Banding.

3.5.1 Band seat.-Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

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3.5.2 Band blank.-The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification, except that where hot banding at not less than 800 degrees Fahrenheit (°F). No prior annealing is necessary.

3.5.3 Application.-The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the sidewall depth. The band shall be applied in such manner as not to distort the body wall. There shall be no visible gap between the rotating band and the sidewalls of the band seat (on the outside surface of the projectile) after the rotating band has been machined to final dimensions.

3.5.3.1 Rotating band.-The bands may be applied either hot or cold but if applied hot, the bands shall not be heated above 1600°F to prepare them for application to the body. The bands shall be heated uniformly by a process which will prevent undue oxidation (loose scale), and applied in a manner that will hold the formation of scale between the band and the band seat to a minimum.

3.5.4 Band seating.-The indicated diametrical clearance between the rotating band and band seat shall not exceed .006 inch.

3.6 Metal defects.

3.6.1 Components (body and adapter).-The components including the cavity shall be free from cracks, splits, burst, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.6.2 Components (body and adapter).-The cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Visual standards.-Visual standards for metal defects will be established for each contract after the first month's production. However, the establishment of visual examination will not waive the intent of the requirements of 3.6.1 and 3.6.2 (see 6.2).

3.6.4 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.7 Proving Ground Functional Test.-The projectile assembly shall reveal no evidence of failure when tested at excess and service pressure. Metal parts failure shall be:

a. Loss or breakup of metal part components in flight.

b. Shear of the rotating band as evidenced by complete lack of engraving.

c. Loose or fanning of the rotating band.

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3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.8.1 Brazed joint.-Braze metal shall be visible for full 360 degrees around exterior of joint between the adapter and body.

3.8.2 Sectionalized joint.-When sectionalized, the braze metal of the joint shall be free of voids, cracks, entrapped slag or any similar defects for a minimum of 85 percent of the aggregate of the exposed joint section. Presence of cleaning or other fluids in the solder groove shall be considered evidence of a defective brazed joint.

3.8.3 Surface preparation.-Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.8.4 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.8.5 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.8.6 Protection of rotating band.-The machined rotating band shall be protected from being nicked, burned and otherwise damaged in subsequent handling prior to shipment.

3.8.7 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.8 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter, prior to inspection and shipping. Particular attention shall be given to the cavity of the metal parts to insure that no grit, scale or other foreign matter enter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground Functional test (see 3.7).

b. Acceptance or rejection of lot by the Government shall be contingent on successful passage of the Proving Ground Functional tests in 4.3.2.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of sixty-five metal parts as follows:

55 Metal Parts Assemblies, complete, including paint
5 Metal Parts Assemblies, complete, except paint
5 Sets of Component Parts, unpainted, unassembled, including rotating band blanks. The bodies shall be completely machined, including the rotating band seat.

Fifty of the above painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions and test methods and procedures.-(see 6.5)

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Bar or Billet soundness (sample).-From each lot of bars or billets supplied in accordance with ASTM Specifications, one bar or billet for each 25,000 pounds of steel or fraction thereof shall be selected for testing from each lot or heat of steel, except that not less than 2 bars or billets and not more than 10 shall represent the lot or heat of steel. Each bar in the sample shall be subjected to the Macro Etch Test to determine compliance with 3.2. Failure of any macro etch specimen to comply shall be cause for rejection of the bar from which the failing specimen was taken, and shall be cause for testing each remaining bar or billet in the lot of heat. Failure of any of the additional specimens to comply with 3.2 shall be cause for rejection of the bar represented by the failing specimen.

4.3.1.3 Bar or Billet soundness (test method).-From the sample bars or billets supplied in accordance with 4.3.1.2, a full cross section specimen approximately one half inch thick shall be removed from each end of each bar or billet. The macro etch test shall be performed in accordance with ASTM Method A-317.

4.3.1.4 Mechanical properties.

4.3.1.4.1 Hardness and tension test.

4.3.1.4.1.1 Heat treatment lots (definition).-Heat treatment lots shall contain heat treated components from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4.1.2 Hardness (heat treatment lots).-In order to determine uniformity of heat treatment, each component in the heat treatment lots (as defined by 4.3.1.4.1.1) shall be subjected to a Brinnell Hardness test or any method of hardness testing approved by the procuring activity.

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4.3.1.4.1.3 Heat treatment lots (tension test samples).-Two components, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Components which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile components which will represent the true extremes of hardness in the heat treatment lot.

4.3.1.4.1.4 Stress relieved lots (definition).-Stress relieved lots shall contain stress relieved components, from not more than one heat of steel, that have been cold shaped by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4.1.5 Hardness (stress-relieved lots).-Twenty-five cold shaped components from the first batch of each stress-relieved lot (as defined by 4.3.1.4.1.4) processed through the stress-relief furnace and five components from each one hour's production thereafter for the balance of the stress relieved lot shall be subjected to a Brinnel hardness test, or any method of hardness testing approved by the procuring activity to assess uniformity of the product.

4.3.1.4.1.6 Stress relieved lots (tension test samples).-Two components, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five components shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout each stress-relief lot. The furnace temperature may vary within twenty-five degrees of Fahrenheit but not below the minimum temperature of 700 degrees Fahrenheit and the minimum 30 minutes soak time. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each eight hours by an independent method.

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4.3.1.4.1.7 Miscellaneous lots (definition).-Small lots (such as components manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress-relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either heat treat or streee-relieved lots as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4.1.8 Hardness (miscellaneous lots).-Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.3.1.4.1.2. Miscellaneous stress-relieved lots shall be subjected to a hardness test in accordance with 4.3.1.4.1.5.

4.3.1.4.1.9 Miscellaneous lot (tension test samples).-Samples for tension test of miscellaneous heat treatment or stress-relieved lots shall be selected in accordance with 4.3.1.4.1.3 or 4.3.1.4.1.6, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirement of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.1.4.1.10 Hardness test methods.-After final heat treatment or stress relief, but prior to finish machining, a Brinnel hardness test shall be performed on the outside surface at the band seat location or at any other area on the body deemed necessary for process control. The equipment and procedure for performing this test are prescribed by ASTM Method E-10, when a Brinnel hardness test is used.

4.3.1.4.1.11 Tension test methods.-From the samples provided by the heat treat, stress relief or miscellaneous lots two tension test specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by ASTM Method E-8, shall be taken longitudinally from the opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

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4.3.1.4.1.12 Bar stock (tension test samples).-When heat treatment is not required during manufacture of components from bar stock, two components from each lot of 1,000 components or fraction thereof shall be subjected to a tension test. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot represented by the sample. The test may be performed prior to final machining.

4.3.1.4.1.13 Bar stock (tension test method).-From the samples supplied above, one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component as prescribed by ASTM Method E-8. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.3.1.4.1.14 Projectile lots.-When it has been determined that components of a heat treated, stress relieved or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned projectile lot number in accordance with MIL-STD-1168. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of a continuous production, free from interruptions other than those due to the end of a shift, day, or work week. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.3.1.4.1.15 Rotating band (tension test samples).-One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subjected to the tension test to determine compliance with 3.3.2. Failure to comply shall be cause for rejection of the production represented by the sample.

4.3.1.4.1.16 Rotating band (tension test method).-The samples provided above shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having one-inch gage length and one-half inch gage width, and grip ends of sufficient length to accomodate the tensile machine shall be removed from the blank and tested in accordance with the procedures prescribed by ASTM Method E-8.

4.3.1.5 Band seating.

4.3.1.5.1 Tightness.

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4.3.1.5.1.1 Non-destructive (samples).-Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with the requirements of the applicable drawing shall be cause for rejection of the production represented by the sample.

4.3.1.5.1.2 Non-destructive method.-The samples provided by 4.3.1.5.1.1 shall be tested utilizing the equipment and procedures prescribed by Frankford Arsenal Detriot Testing Machine Manual (see 6.3).

4.3.1.5.1.3 Destructive (samples).-When the non-destructive equipment is not available, one body assembly from each 8-hour production or fraction thereof from each banding machine in operation shall be subjected to this test to determine compliance with 3.5.4. Failure to comply shall be cause for rejection of the production represented by the sample.

4.3.1.5.1.4 Destructive method.-The body assembly selected above should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.3.1.5.1.5 Sidewall contact.-The samples provided for the tension test shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of rotating band. The cylindrical portion containing the rotation band shall be sectioned approximately in half through a plane parellel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other half shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surface shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.5.3.

4.3.1.6 Brazed joint.

4.3.1.6.1 Hydrostatic test.-Each metal parts assembly shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

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4.3.1.6.2 Hydrostatic pressure testing.-This test will be performed using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure has been reached, after which the pressure will be maintained for the minimum prescribed time. The equipment shall be calibrated prior to the start of each day's operatiion, and at intervals of not more than 4 hours of operation. The equipment shall also be calibrated prior to restarting the operation after a discontinuance of one hour or more. After acceptance stamp letter "H" as indicated on assembly drawing.

4.3.1.6.3 Air pressure test.-After hydrostatic test and drying time, each metal parts assembly shall be subjected to the air test to determine compliance with the requirements of the applicable drawing. Failure of any assembly to comply, shall be cause for rejection of the assembly.

4.3.1.6.4 Air pressure test (method).-This test will be performed using equipment capable of performing the test accurately and consistently. The equipment shall be calibrated prior to the start of each day's operation and after each four (4) hours continous operation. The assembly shall be submerged in water in such a manner that the nose adapter joint is covered and observation can be made for the full 360 degress of the adapter joint. The pressure will be applied until the required minimum pressure is reached after which, the pressure will be held for the minimum time specified on the applicable drawing. Observation shall be made for air bubbles rising the surface of the water or bubbles leaving the vicinity of the adapter joint for the entire time required. After acceptance, stamp letter "A" as indicated on the applicable drawing.

4.3.1.6.5 Joint quality.-After hydrostatic and air test and prior to application of protective coating two projectile assemblies from each lot shall be selected at random, sectioned and separated, to determine compliance with 3.8.6. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.1.6.6 Joint quality.-The samples selected in accordance with 4.3.1.6.5 shall be sectioned and separated as follows:

a. The forward portion of the projectile assembly, 1-inch rear of the braze joint shall be removed in such a manner that no portion of the braze will be damaged.

b. The forward portion of the body containing the adapter shall be cut vertically through the center of the adapter.

c. Each portion of the body shall be peeled from the adapter and the mating pieces of the body shall be visually examined to determine compliance with the requirements.

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4.3.1.7 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the action recommended by the applicable specification.

4.3.1.7.1 Salt spray.-The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure to comply with the requirements of the applicable drawing shall be cause for taking corrective action recommended by the applicable specification. The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by ASTM B-117.

4.3.1.7.2 Protective coating.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification.

4.3.2 Proving Ground functional tests.-The following Proving Ground tests will be performed by the Government.

4.3.2.1 First Article sample.

4.3.2.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	10
Service pressure	10

4.3.2.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
*Excess pressure	10

* After three (3) consecutive production lots have been tested successfully, the excess pressure phase may be discontinued, with the approval of the procuring agency. However, implementation of the excess pressure phase shall be instituted whenever directed by the procuring agency.

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4.3.2.3 Excess pressure.-The projectiles supplied in accordance with Tables I and II, will be fired for recovery, in a gun for which the projectile is standard, the tube of which will have 50 percent life remaining, based on current gun tube condemnation criteria. The propellant charge will be adjusted to obtain 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile, for the initial production sample and the first three (3) production lot samples fired. An attempt will be made to recover at least 60 percent of the projectiles fired in this phase for examination to determine compliance with 3.7. Failure to comply shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.2.4 Service pressure.-The projectiles supplied in accordance with Tables I and II, will be fired alternately with projectiles from a reference lot for comparison. The projectile shall be fired in a gun for which the round is standard, the tube of which shall have a minimum remaining service life of 50 percent, based upon currect gun tube condemnation criteria. The projectile will be fired for accuracy at a range selected by the proof director. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.7. Failure to comply shall be cause for rejection of the initial production sample represented by the sample.

4.3.3 Inspection equipment.-Index of Inspection Equipment List No. LI-10535390 identifies the inspection equipment required to perform the examination and tests prescribed herein. Reference is made to the LI for definition of coding format.

4.3.4 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity (see 6.6). All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual inspection. The roughness comparison specimens prescribed by ANSI B46.1 shall be used as a basis for surface roughness determination.

4.3.4.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during first article inspection and after each 250 items produced for each machine in operation. Where destruction of components is necessary to inspect these dimensions, measurement of the tool may be substituted providing the contractor has established correlation between tool dimensions and component dimensions prior to start of production. The contractor shall be responsible for maintaining adequate surveillance to assure drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

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4.3.4.2 Adapter, Drawing D10535944, covering a detail of Drawing F10535941.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter of brazing surface	Gage
MINOR	
201. Length to forward basic diameter on rear taper	Gage
202. Length to rear basic diameter on rear taper	Gage
203. Length from front face to shoulder	Gage
204. Length of brazing surface	Gage
205. Length, total	Gage
206. Presence and dimension of solder ring groove (if applicable)	Visual (see 6.2)
207. Presence and dimensions of undercut (if applicable)	Visual (see 6.2)
208. Radii or chamfers missing or incorrect	Visual (see 6.2)
209. Surface finish improper	Visual (see 6.2)
210. Burr	Visual
211. Foreign matter	Visual

4.3.4.3 Body, Drawing F10534942, covering a detail of Drawing F10534941.

CRITICAL	
1. Metal defective	Visual (Probe light)
MAJOR	
101. Runout of brazing surface with bourrelet	Gage
102. True position of basic diameter in cavity recess with brazing diameter	Gage
103. Diameter of brazing surface	Gage
104. Diameter of rotating band seat	Gage
105. Diameter of cavity recess flat, min.	Gage
106. Depth from front face to bottom of cavity recess	Gage
107. Length of brazing surface, min.	Gage
108. Length to forward ogive basic diameter	Gage

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4.3.4.3 (Cont'd)

Defect	Method of Inspection
109. Length to rear ogive basic diameter	Gage
110. Length from bottom of recess cavity to basic diameter in recess cavity	Gage
111. Length from front face to rear of forward bourrelet	Gage
112. Location of rotating band seat	Gage
113. Width of rotating band seat	Gage
114. Variation in wall thickness	Gage
115. Foreign matter on rotating band seat	Visual
MINOR	
201. Diameter of chamfer of brazing surface	Gage
202. Length, total	Gage
203. Radii or chamfers missing or incorrect	Visual (see 6.2)
204. Surface finish improper	Visual (see 6.2)
205. Scoring missing from rotating band seat	Visual
206. Evidence of shot or sandblasting on rotating band seat	Visual
207. Burr	Visual
208. Foreign matter (except as otherwise classified)	Visual

4.3.4.4 Body Assembly, Drawing F10535941, covering a detail of Drawing D10535930 (Before paint).

CRITICAL	
1. Braze metal not visible around the entire periphery of the body adapter joint	Visual
2. Metal defective	Visual (Probe light)
MAJOR	
101. Weight	Approved scale
102. Runout of pitch diameter of thread with bourrelets and front face	Gage
103. Runout of counterbore below fuze cavity with bourrelets	Gage
104. Runout of press fit diameter with bourrelets	Gage
105. Runout of adapter with bourrelets	Gage
106. Runout of body ogive with bourrelets	Gage

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4.3.4.4 (Cont'd)

Defect	Method of Inspection
107. Runout of body with bourrelets	Gage
108. Runout of rotating band with bourrelets	Gage
109. Runout of boattail with bourrelets	Gage
110. Diameter of bourrelets, min.	Gage
111. Diameter of body	Gage
112. Diameter of rotating band flat	Gage
113. Length to forward boattail basic diameter	Gage
114. Length to rear boattail basic diameter	Gage
115. Length to forward basic diameter of adapter	Gage
116. Length to rear basic diameter of adapter	Gage
117. Length, total, max.	Gage
118. Location of rotating band	Gage
119. Width of rotating band	Gage
120. Width at front of rotating band, incorrect	Gage
121. Width of rotating band flat	Gage
MINOR	
201. Diameter of rotating band relief groove	Gage
202. Angle at rear of rotating band relief groove	Gage
203. Length of rotating band relief groove flat	Gage
204. Surface finish improper	Visual (see 6.2)
205. Radii or chamfers missing or incorrect	Visual (see 6.2)
206. Presence and dimension of undercut forward of rotating band	Visual (see 6.2)
207. Burr	Visual
208. Foreign matter	Visual

4.3.4.5 Metal Parts Assembly, Drawing D10535930.

CRITICAL	
1. Diameter of press fit area	Gage
2. Reverse taper in press fit area	Gage
3. Surface finish of press fit area	Gage
4. Profile of radius with press fit diameter for 360 degrees	Visual (see 6.2)
5. Hydrostatic test stamp missing (indicating failure to pass hydrostatic test)	Visual
6. Air pressure stamp missing (indicating failure to pass air test)	Visual
7. Metal defective	Visual (Probe light)

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4.3.4.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Pitch diameter of thread, min.	Gage
103. Minor diameter of thread, max.	Gage
104. Diameter of bourrelet, max.	Gage
105. Diameter of counterbore at rear of fuze cavity	Gage
106. Depth to bottom of fuze cavity	Gage
107. Depth to bottom of counterbore at rear of fuze cavity	Gage
108. Length of effective thread, min.	Gage
109. Rotating band damaged	Visual
110. VCI crystals or tablets missing	Visual
MINOR	
201. Diameter of thread chamfer	Gage
202. Length of thread, max.	Gage
203. Protective finish, inadequate or damaged	Visual
204. Marking missing, incorrect or illegible	Visual
205. Paint on threads	Visual
206. Paint on front face of adapter	Visual
207. Paint on rotating band bearing surfaces	Visual
208. Paint not dry	Visual
209. Corrosive preventive compound inadequate or excessive	Visual

4.3.4.6 Pallet, Drawing F8837835.

CRITICAL

None defined

MAJOR

None defined

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4.3.4.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve(s) missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual and Manual
205. Marking missing, incorrect or illegible	Visual

5. PREPARATION FOR DELIVERY

5.1 Level C.-Minimum military pack.

5.1.1 Packaging and packing.-Packaging, packing and marking shall be in accordance with MIL-STD-1169 and the following:

5.1.1.1 Projectile metal parts assemblies, will be palletized in accordance with Drawing F8837835.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractor's plant in compliance with 3.5.3 or whenever directed by the procuring agency.

6.3 Band seating.-The operating procedures for use of Detroit Testing Machine as prepared by the Quality Branch, US Army, Frankford Arsenal, SARFA-QAA-Q.

6.4 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.5 Alternative testing procedures.-Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisons specified herein. Prior to applying the alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or better than the quality assurance provisions specified herein.

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6.6 When product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, product produced prior to the discovered defect will be screened until the frequency (F) is found to be free of that defect.

Custodian:

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Preparing Activity:

Army - MU

Project No. 1315-A254

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